



Python Programming - 2301CS404

Lab - 14

Roll No.:315

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Import matplotlib below

```
In [4]: import matplotlib.pyplot as plt
```

01) WAP to display the line chart for given data for x and y axis.

```
In [50]: x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

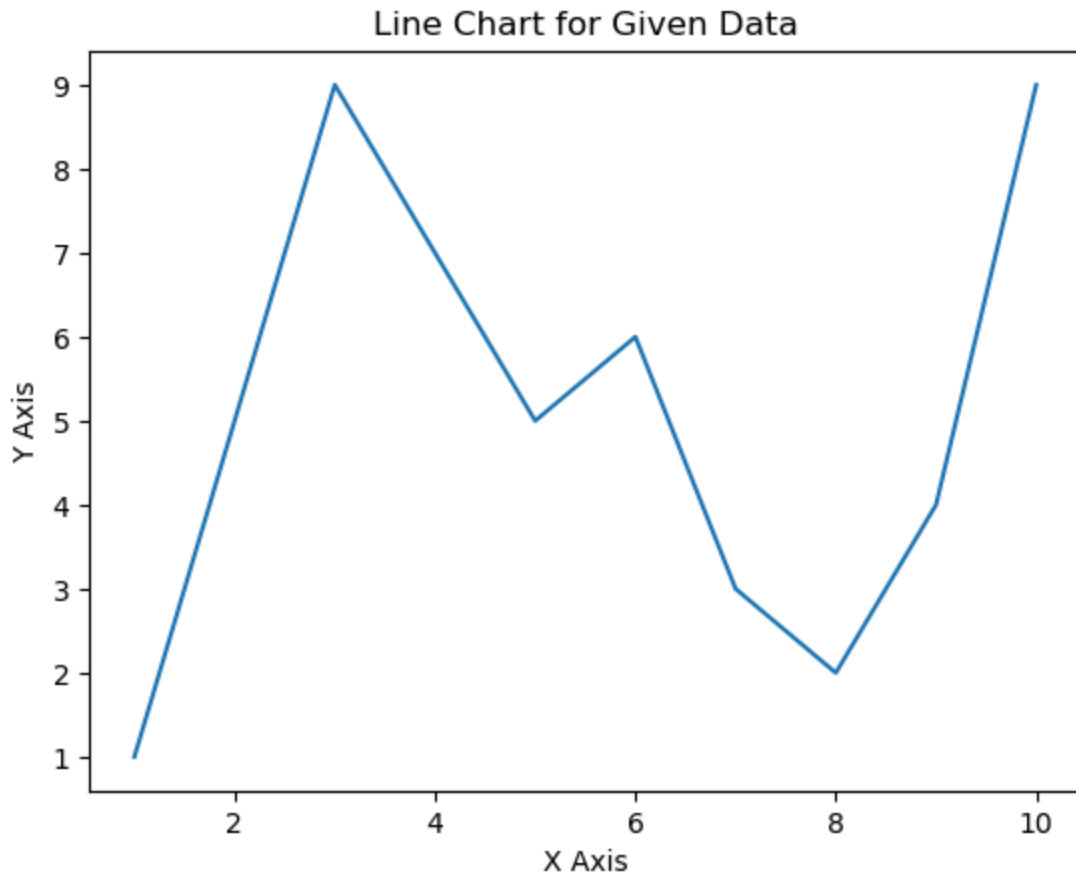
```
In [51]: x = range(1, 11)
y = [1, 5, 9, 7, 5, 6, 3, 2, 4, 9]

plt.plot(x, y)

plt.xlabel("X Axis")
plt.ylabel("Y Axis")

plt.title("Line Chart for Given Data")

plt.show()
```



02) WAP to generate line chart for given data.

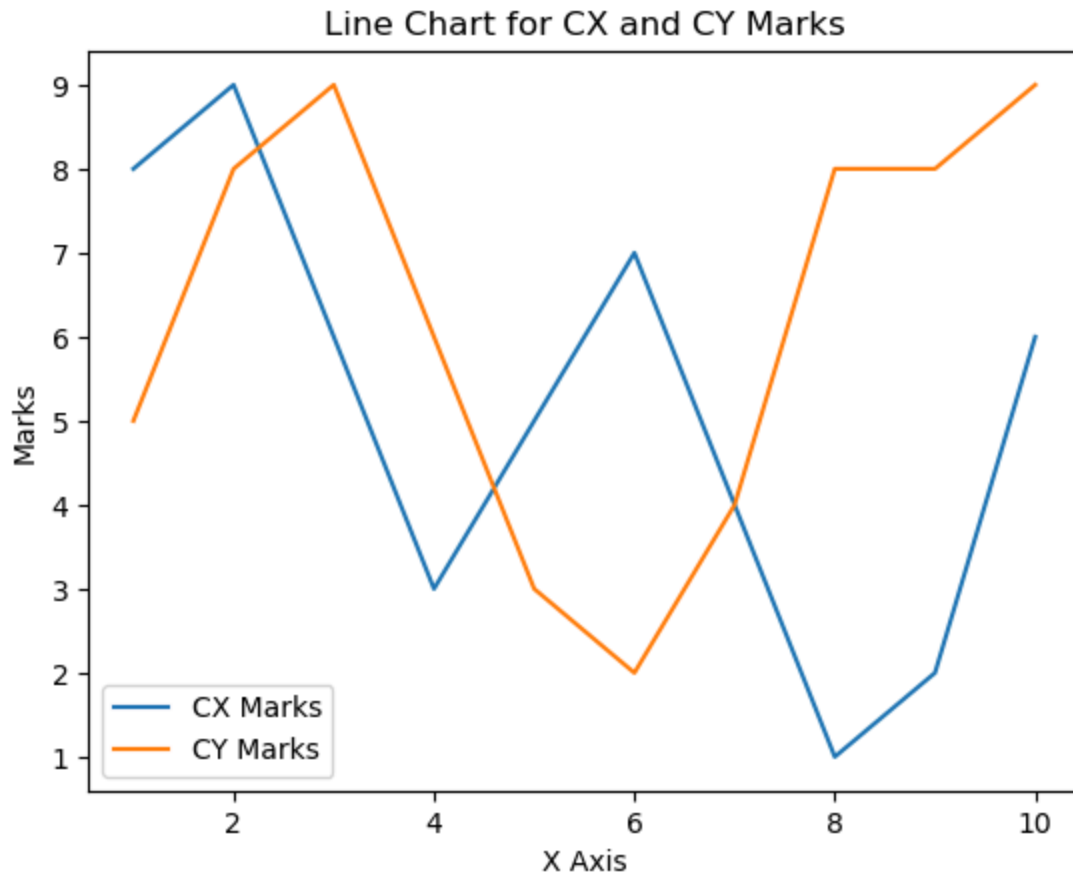
```
In [5]: x = range(1,11,1)
cxMarks= [8,9,6,3,5,7,4,1,2,6]
cyMarks= [5,8,9,6,3,2,4,8,8,9]
```

```
In [6]: plt.plot(x, cxMarks, label="CX Marks")
plt.plot(x, cyMarks, label="CY Marks")

plt.xlabel("X Axis")
plt.ylabel("Marks")
plt.title("Line Chart for CX and CY Marks")

plt.legend()

plt.show()
```



03) WAP to generate line chart for given data, but create your custom axis.

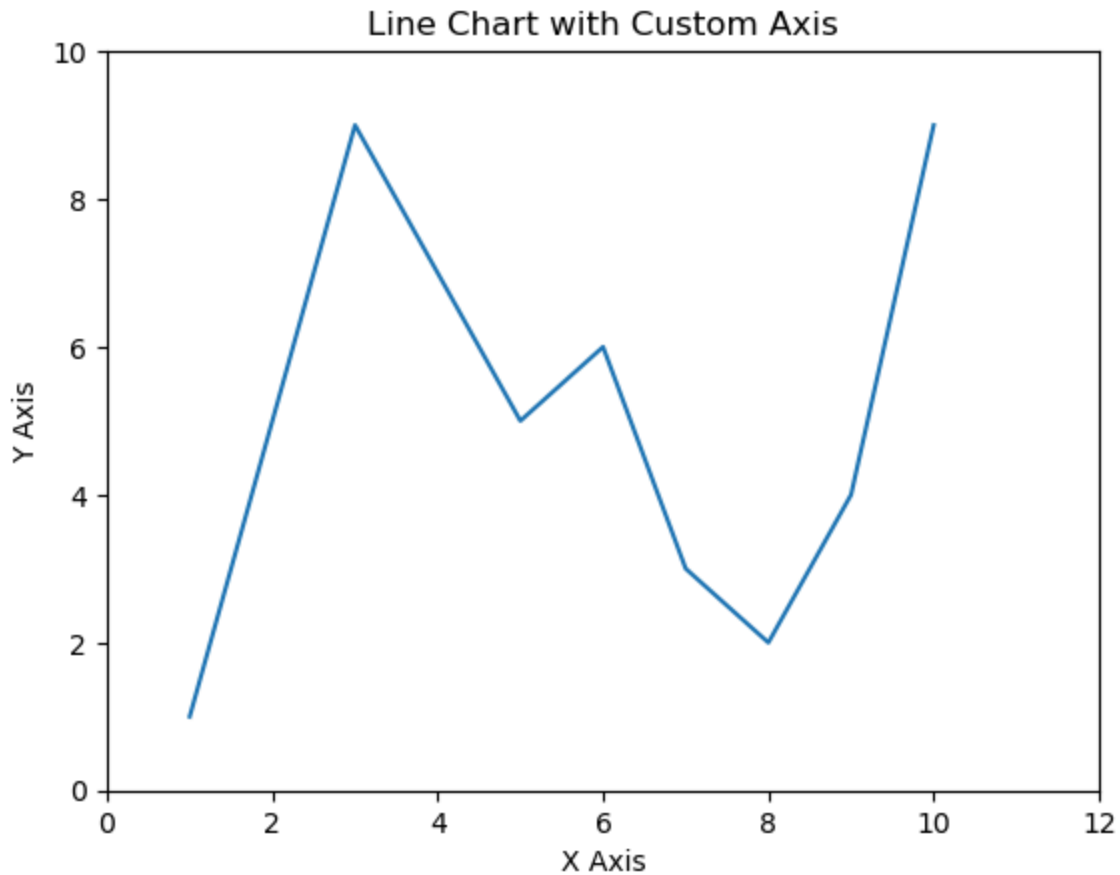
```
In [9]: x = range(1,11)
        y = [1,5,9,7,5,6,3,2,4,9]
```

```
In [14]: # Plot line chart
plt.plot(x, y)

# Custom axis range
plt.axis([0, 12, 0, 10]) # [xmin, xmax, ymin, ymax]

# Labels and title
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Line Chart with Custom Axis")

# Display chart
plt.show()
```



04) WAP to plot a line chart showing temperature variation during a week. Mark the lowest and highest temperature.

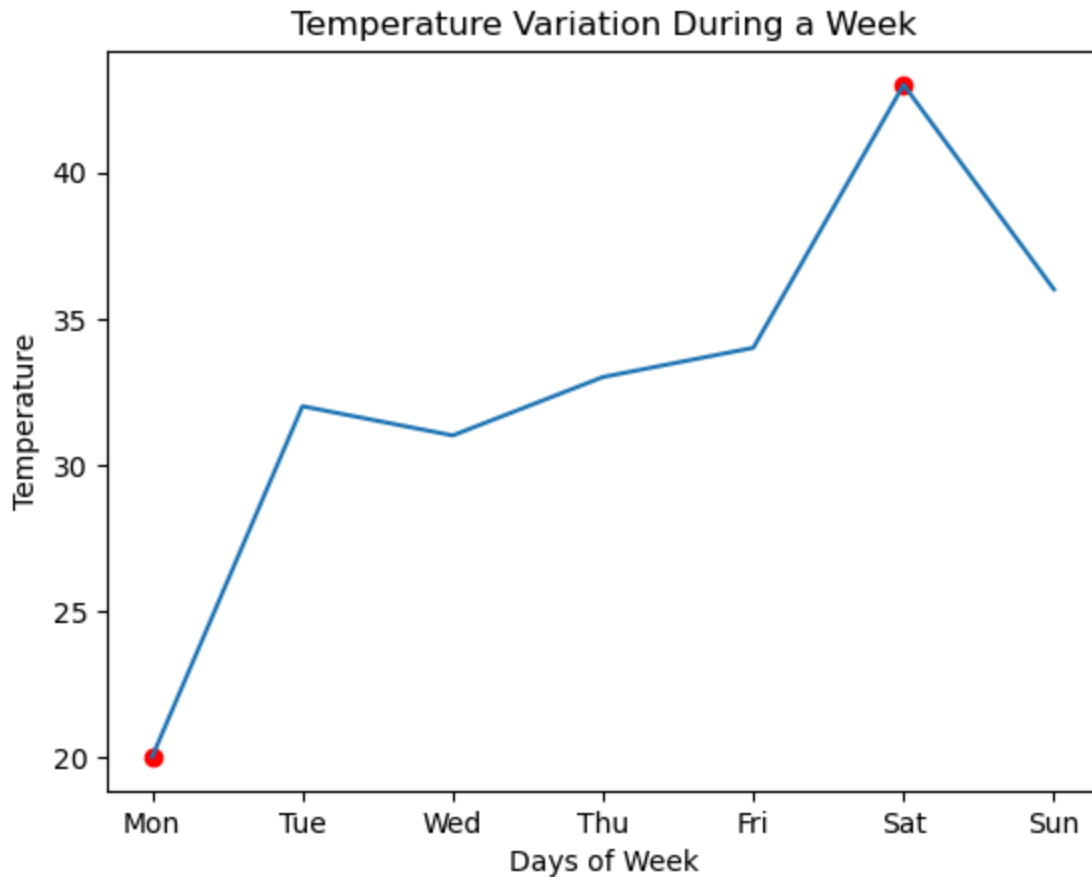
```
In [16]: Days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
         Temperature = [20, 32, 31, 33, 34, 43, 36]
```

```
In [23]: # Plot line chart
         plt.plot(Days, Temperature)

         plt.scatter(Days[Temperature.index(max(Temperature))], max(Temperature), c='red')
         plt.scatter(Days[Temperature.index(min(Temperature))], min(Temperature), c='red')

         # Labels and title
         plt.xlabel("Days of Week")
         plt.ylabel("Temperature")
         plt.title("Temperature Variation During a Week")

         # Show chart
         plt.show()
```



05) WAP to create a detailed pie chart representing the monthly expenditure distribution of a family.

The chart should display labels, percentage values, title, different colors for each slice, and explode one slice to highlight the highest expense.

```
In [26]: Categories = ["Rent", "Food", "Education", "Transport", "Entertainment", "Savings"]
Amount = [15000, 8000, 6000, 3000, 2000, 5000]
```

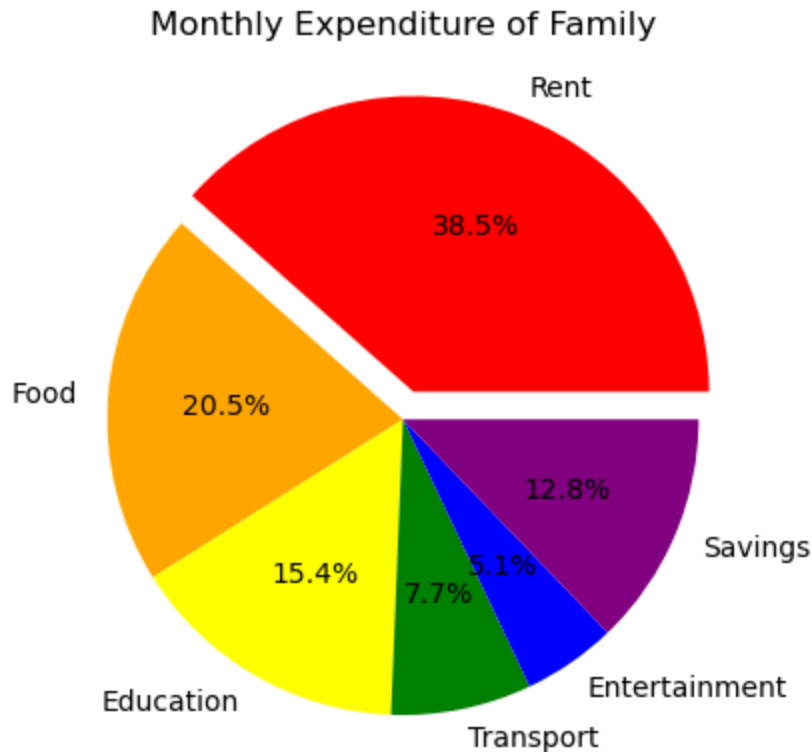
```
In [27]: # explode the highest expense (Rent)
explode = [0.1, 0, 0, 0, 0, 0]

colors = ["red", "orange", "yellow", "green", "blue", "purple"]

plt.pie(Amount, labels=Categories, autopct='%1.1f%%',
        colors=colors, explode=explode)

plt.title("Monthly Expenditure of Family")

plt.show()
```



06) WAP to create bar chart showing the marks obtained by students in different subjects.

The program should also display the marks on the top of the respective bar.

```
In [2]: Subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]
Marks = [78, 85, 72, 80, 90]
```

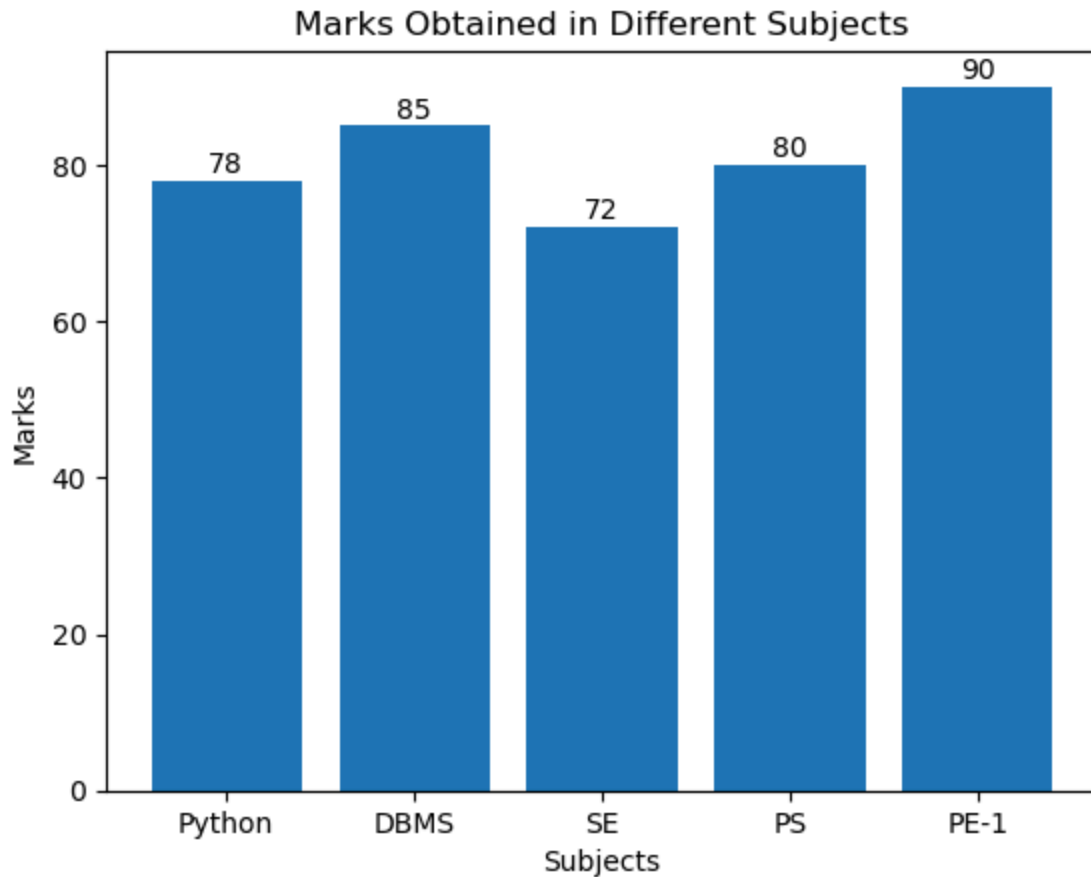
```
In [8]: subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]
marks = [78, 85, 72, 80, 90]

bars = plt.bar(subjects, marks)

plt.title("Marks Obtained in Different Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks")

for i in range(len(subjects)):
    plt.text(i, marks[i] + 1, marks[i], ha='center')

plt.show()
```

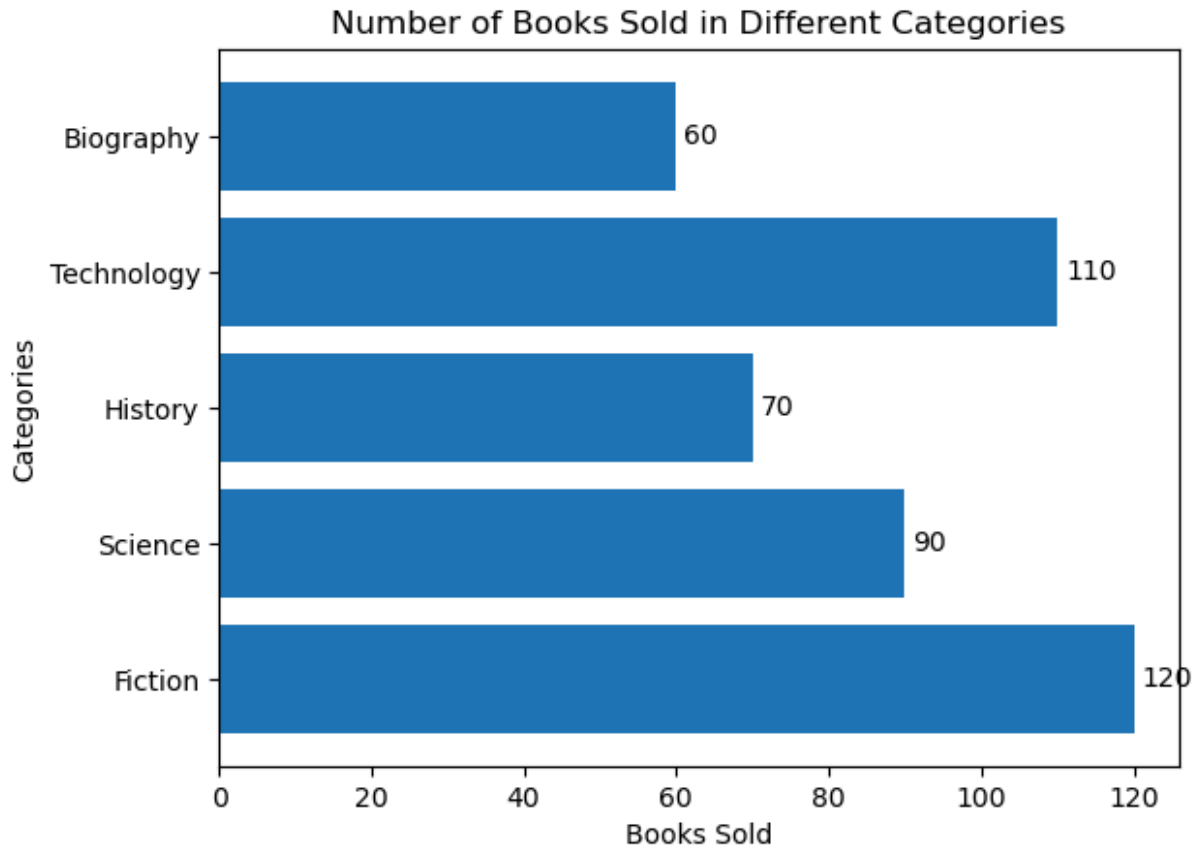


07) WAP to create a horizontal bar chart representing the number of books sold in different categories.

The program should also display the number of books sold at the end of the respective bar.

```
In [ ]: Categories = ["Fiction", "Science", "History", "Technology", "Biography"]  
Books Sold = [120, 90, 70, 110, 60]
```

```
In [7]: categories = ["Fiction", "Science", "History", "Technology", "Biography"]  
books_sold = [120, 90, 70, 110, 60]  
  
plt.barh(categories, books_sold)  
  
plt.title("Number of Books Sold in Different Categories")  
plt.xlabel("Books Sold")  
plt.ylabel("Categories")  
  
for i in range(len(categories)):  
    plt.text(books_sold[i] + 1, i, str(books_sold[i]), va='center')  
  
plt.show()
```



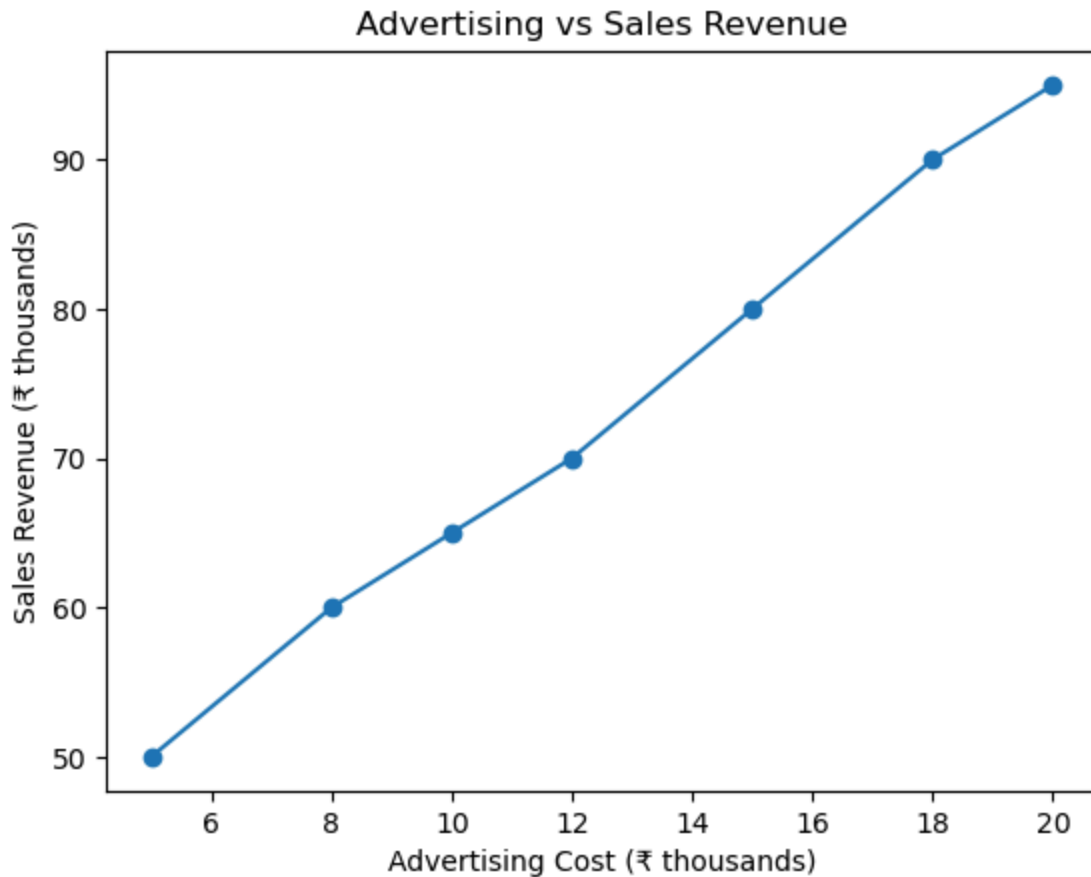
08) WAP to create a scatter plot representing the relationship between advertising expenditure and sales revenue of a company.

```
In [57]: Advertising_Cost = [5, 8, 10, 12, 15, 18, 20]
Sales_Revenue = [50, 60, 65, 70, 80, 90, 95]
```

```
In [59]: plt.plot(Advertising_Cost, Sales_Revenue)
plt.scatter(Advertising_Cost, Sales_Revenue)

plt.xlabel("Advertising Cost (₹ thousands)")
plt.ylabel("Sales Revenue (₹ thousands)")
plt.title("Advertising vs Sales Revenue")

plt.show()
```

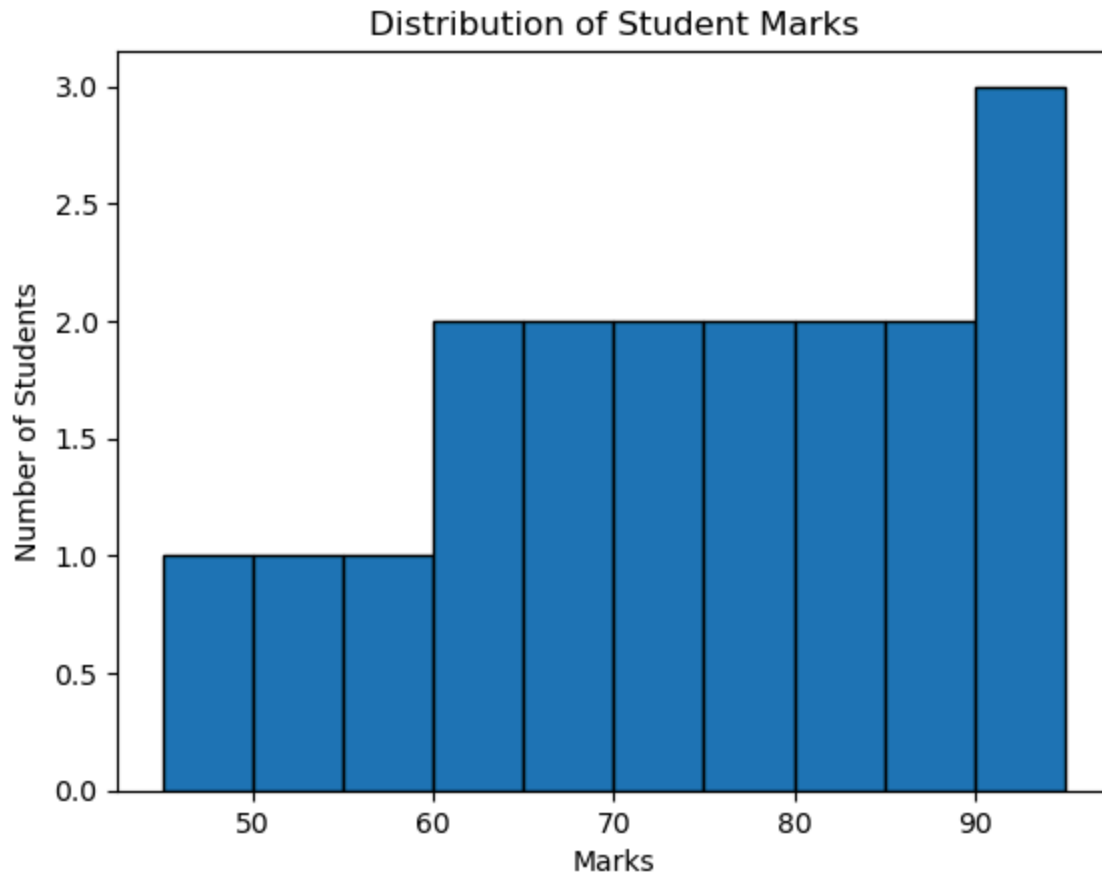
09) WAP to create a histogram representing the distribution of marks obtained by students in a class.

```
In [60]: Marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95]
```

```
In [67]: plt.hist(Marks , edgecolor="black")

plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")

plt.show()
```



10) WAP to create a box plot representing the distribution of daily commuting time (in minutes) of employees and identify outliers (if any).

```
In [ ]: Commuting Time (minutes) = [5, 18, 20, 22, 25, 27, 28, 30, 32, 35, 38, 40, 42, 90]
```

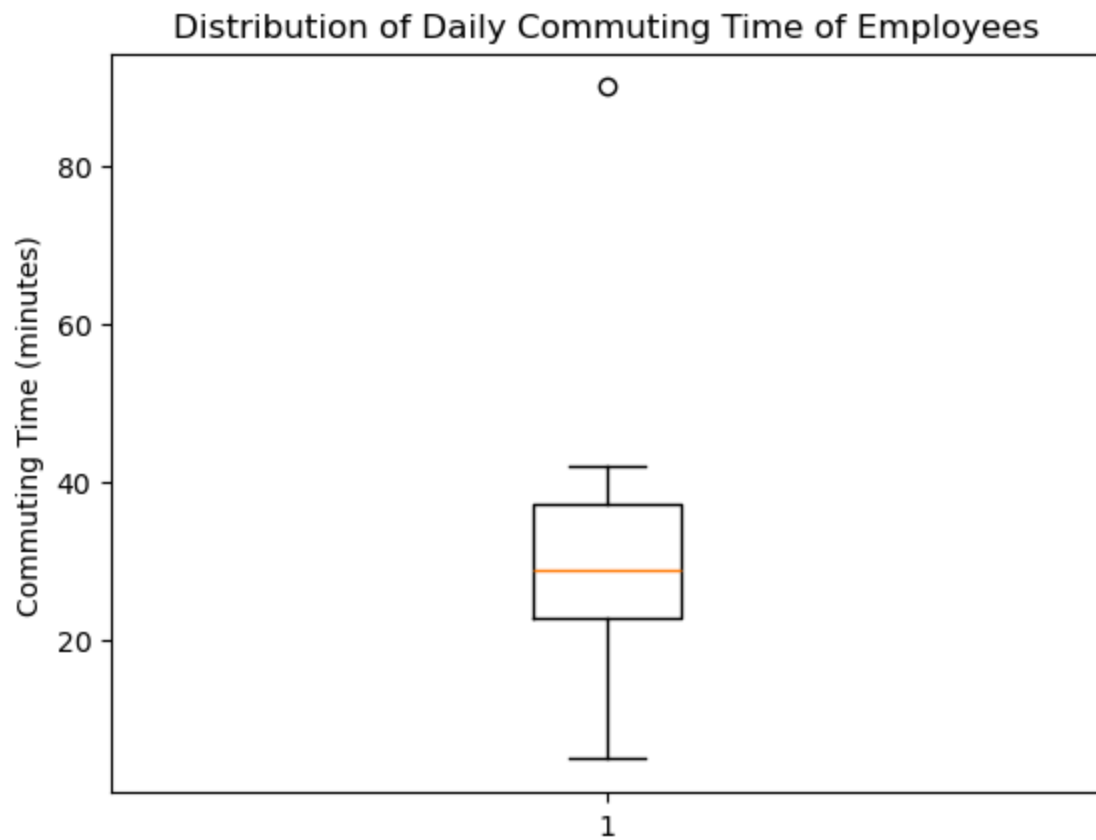
```
In [6]: commuting_time = [5, 18, 20, 22, 25, 27, 28, 30, 32, 35, 38, 40, 42, 90]
```

```
plt.boxplot(commuting_time)
```

```
plt.title("Distribution of Daily Commuting Time of Employees")
```

```
plt.ylabel("Commuting Time (minutes)")
```

```
plt.show()
```



In [33]: %ls

Volume in drive D is New Volume
Volume Serial Number is 946D-FE5D

Directory of D:\B_Tech_SEM_4\PYP\PYTHON_lab

```

10/03/2026  10:40 AM    <DIR>          .
26/02/2026  12:53 AM    <DIR>          ..
17/02/2026  09:57 AM    <DIR>          .ipynb_checkpoints
10/02/2026  10:19 AM                126 abc.txt
30/11/2025  04:44 PM    <DIR>          anaconda_projects
25/11/2025  11:00 AM                33,495 FIRSTHELLO.pdf
10/02/2026  11:07 AM                41 input.txt
17/02/2026  11:08 AM               318,530 lab_12_pdf.pdf
02/12/2025  06:57 PM               13,615 Lab_2.ipynb
02/12/2025  07:00 PM               297,080 LAB_2_pdf.pdf
09/12/2025  11:04 AM               292,514 LAB_3_pdf.pdf
23/12/2025  05:28 PM               355,732 Lab_5.pdf
13/01/2026  11:29 AM               403,064 Lab_8.pdf
20/01/2026  11:22 AM                8,500 Lab_9.ipynb
20/01/2026  11:21 AM               260,653 Lab_9_pdf.pdf
03/02/2026  11:11 AM               319,489 lab-10-pdf.pdf
10/02/2026  11:19 AM               314,080 lab-11.pdf
30/12/2025  11:27 AM               335,425 Lab-6.pdf
03/02/2026  11:01 AM                20 marks.txt
03/02/2026  11:03 AM               214 merged.txt
03/02/2026  11:06 AM               107 new.txt
10/02/2026  10:45 AM                22 numbers.txt
10/02/2026  11:18 AM               728 output.txt
03/02/2026  11:02 AM               471 primenumbers.txt
03/02/2026  11:24 AM               12,027 Python Programming - Lab - 10.ipynb
10/02/2026  11:17 AM               11,264 Python Programming - Lab - 11.ipynb
17/02/2026  11:07 AM               14,928 Python Programming - Lab - 12.ipynb
10/03/2026  10:40 AM              223,381 Python Programming - Lab - 14.ipynb
09/12/2025  11:03 AM               11,229 Python Programming - Lab - 3.ipynb
23/12/2025  05:28 PM               15,258 Python Programming - Lab - 5.ipynb
30/12/2025  11:26 AM               16,228 Python Programming - Lab - 6.ipynb
06/01/2026  11:28 AM               14,870 Python Programming - Lab - 7.ipynb
13/01/2026  11:28 AM               16,281 Python Programming - Lab - 8.ipynb
27/01/2026  11:28 AM               13,861 Python Programming - Lab - 9.ipynb
27/01/2026  11:20 AM              364,430 Python Programming - Lab - 9.pdf
10/02/2026  11:11 AM                46 stu.txt
03/02/2026  10:04 AM                0 untitled.txt
          33 File(s)          3,667,709 bytes
          4 Dir(s)   171,320,168,448 bytes free

```

In [34]: `%pwd`

Out[34]: 'D:\\B_Tech_SEM_4\\PYP\\PYTHON_lab'

In [39]: `%cd "Asus"`

[WinError 2] The system cannot find the file specified: 'Asus'
C:\Users\Asus

In [40]: `%time`

CPU times: total: 0 ns

Wall time: 11.4 µs

In [44]: %whos

Variable	Type	Data/Info
dataframe_columns	function	<function dataframe_colum<...>ns at 0x000001D571C97CE0>
dataframe_hash	function	<function dataframe_hash at 0x000001D571C960C0>
dtypes_str	function	<function dtypes_str at 0x000001D571C96020>
get_dataframes	function	<function get_dataframes at 0x000001D571C971A0>
getpass	module	<module 'getpass' from 'C<...>conda3\\Lib\\getpas s.py'>
hashlib	module	<module 'hashlib' from 'C<...>conda3\\Lib\\hashli b.py'>
import_pandas_safely	function	<function import_pandas_s<...>ly at 0x000001D571C95D00>
is_data_frame	function	<function is_data_frame at 0x000001D571C97D80>
json	module	<module 'json' from 'C:\\<...>\\Lib\\json__init_ _.py'>

In [42]: %reset

In []: